Multi-Server Formulas

As can be seen below, the formulas for multi-server performance measures are quite formidable, particularly for large values of S. Fortunately, there are easier methods for obtaining values: we can use a) tables or b) count the formulas in spreadsheets. The best idea is - of course - use a professional software which is available on the software market.

Following formulas are used to measure performance of the multiple-server systems:

1. System utilisation

$$r = \frac{\lambda}{S\mu}$$

2. Probability of n units in the system, where n [illegible] S

$$p_n = p_0 \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}$$

3. Probability of n units in the system, where n > S

$$p_n = p_0 \frac{\left(\frac{\lambda}{\mu}\right)^n}{s!(s^{n-s})}$$

4. Probability of zero units in the system

$$p_0 = \left[\sum_{n=0}^{s-1} \left(\frac{\lambda}{\mu}\right)^n + \frac{\left(\frac{\lambda}{\mu}\right)^s}{s!(1 - \frac{\lambda}{S\mu}} \right]^{-1}$$

5. Probability, that an arrival will have to wait for service

$$p_w = \left(\frac{\lambda}{\mu}\right)^s \frac{p_0}{s!(1 - \frac{\lambda}{S\mu})}$$

6. Average number in line

$$L_q = \frac{\lambda\mu\left(\frac{\lambda}{\mu}\right)^s}{(s-1)!\,(S\mu-\lambda)^2}\; p_0$$

7. Average number in the system

$$L = L_q + \frac{\lambda}{\mu}$$

8. Average time in line

$$W_q = \frac{L_q}{\lambda}$$

9. Average time in the system

$$W = W_q + \frac{1}{\mu}$$

Example

Management of a shop plans to open during Sundays. The shop will have three cash registers with an expected service time 1 minute, projected mean arrival rate is expected to be 1,2 customers per minute.

We shall compute basic performance measures using formulas for multiple-server model.

First, λ = 1,2; μ = 1,0; S = 3. The units of time will be minutes.

The system utilisation

$$r = \frac{1{,}2}{3\cdot 1{,}0} = 0{,}4$$.

Other formulas require the probability p_0; according to formula 4 we get

$$p_0 = \frac{1}{\left[\frac{\left(\frac{1{,}2}{1}\right)^0}{0!} + \frac{\left(\frac{1{,}2}{1}\right)^1}{1!} + \frac{\left(\frac{1{,}2}{1}\right)^2}{2!}\right] + \frac{\frac{\left(\frac{1{,}2}{1}\right)^3}{0!}}{3!\left(1-\frac{1{,}2}{3\cdot 1}\right)}} = 0{,}294$$

Using p_0 we can easy determine other measures:

$$L_q = \frac{1{,}2\cdot 1\left(\frac{1{,}2}{1}\right)^3}{(3-1)!\,(3\cdot 1-1{,}2)^2}\,p_0 = 0{,}32\cdot p_0 = 0{,}32\cdot 0{,}294 = 0{,}094$$ customers in the line.

$$L = 0{,}94 + \frac{1{,}2}{1} = 1{,}294$$ customers in the system.

$$W_q = \frac{0{,}{,}94}{1{,}2} = 0{,}078$$ is average time a customer spends in the line.

$$W = 0{,}78 + \frac{1}{1} = 1{,}078$$ is average time a customer spends in the system.

When will a customer have to wait? The probability of this event must be P_w calculated for S = 3:

$$P_w = \left(\frac{1{,}2}{1}\right)^3 \frac{0{,}294}{3!\left(1 - \frac{1{,}2}{3 \cdot 1}\right)} = 0{,}141$$.

We can conclude: practically no queue exists in the system, the probability that a customer will wait is very small. System should run well.

Of course, the next step of the solution is to prove if would be possible to decrease the number of cash registers to two.